5 (1) $\frac{1}{n}\sin n\theta$

$$\lim_{n\to\infty}\frac{1}{n}\sin n\theta = \lim_{n\to\infty}\frac{\sin n\theta}{n}$$

$$= \lim_{n\to\infty}\frac{\sin n\theta}{n\theta}\cdot\theta$$

$$= \theta$$

(8) $\sqrt{n^2+6n+1} - n$

$$\lim_{n\to\infty}\sqrt{n^2+6n+1} - n$$

$$= \lim_{n\to\infty}\frac{6n+1}{\sqrt{n^2+6n+1}+n} = \lim_{n\to\infty}\frac{6+\frac{1}{n}}{\sqrt{1+\frac{6}{n}+\frac{1}{n^2}}+1}$$

$$= 3$$

(10) $n\left(\sqrt{1+\frac{1}{n}}-1\right)$

$$\lim_{n\to\infty} n\left(\sqrt{1+\frac{1}{n}}-1\right)$$

$$= \lim_{n\to\infty}\frac{\sqrt{1+\frac{1}{n}}-1}{\frac{1}{n}}$$

$$= \lim_{n\to\infty} n\cdot\frac{\frac{1}{n}}{\sqrt{1+\frac{1}{n}}+1} = \lim_{n\to\infty}\frac{1}{\sqrt{1+\frac{1}{n}}+1}$$

$$= \frac{1}{2}$$ ▨

(13) $\sum_{k=1}^{n}\frac{1}{k(k+1)}$

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{k(k+1)} = \lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{k}-\frac{1}{k+1}$$

$$= 1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots$$

$$= 1$$ ▨

✓ 6 (2)

$$\lim_{n\to\infty}\sqrt[n]{n} = \lim_{n\to\infty} n^{\frac{1}{n}} = \alpha$$

$$\lim_{n\to\infty}\frac{1}{n}\ln n = \ln\alpha$$

$$\lim_{n\to\infty}\frac{\ln n}{n} = 0 = \ln\alpha$$

$\therefore \alpha = 1$ "

$\therefore \lim_{n\to\infty}\sqrt[n]{n} = 1$ 이 성립

✓ 9. $n \geq 9$ 에서 $2^n > (n+1)!$ 이 성립한다.

$\therefore \frac{1}{2^n} < \frac{1}{(n+1)!}$ 이 성립 "

$a_n - a_{n+1} = \frac{1}{(n+1)!}$ $(n\geq 2)$ 이고

$\sum_{n=2}^{\infty} a_n - a_{n+1} = \sum_{n=2}^{\infty}\frac{1}{(n+1)!} < \sum_{n=2}^{\infty}\frac{1}{2^n}$ = 수렴 "

$\therefore a_n$은 수렴한다. "

11 $a_n > \frac{3\cdot 2^n - 1}{2^{n+1}+1}$, $a_n < b_n + \frac{3}{2}$ "

$\frac{3\cdot 2^n - 1}{2^{n+1}+1} < a_n < b_n + \frac{3}{2}$ "

$\lim_{n\to\infty}\frac{3\cdot 2^n-1}{2^{n+1}+1} \leq \lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n + \frac{3}{2}$ "

$\therefore \lim_{n\to\infty} a_n = \frac{3}{2}$ ▨